Binary System

Representing negative numbers

① Sign magnitude

1	0	1	1	0

1 ~ negative
0 ~ positive

Magnitude

① 1's compliment

Assume a 4-bit word

+ve
+6 = 0 1 1 0
↓ inverse
-ve
-6 = 1 0 0 1

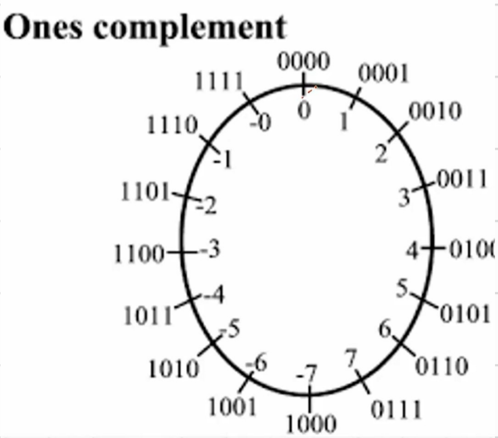
+ve
+7 = 0 1 1 1
↓ inverse
-ve
-7 = 1 0 0 0

+8 = 1 0 0 0 ✗

For a 4-bit word in 1's complimentary, the leftmost bit must be used to express sign not magnitude.

∴ Since > 8 needs to use 4-bits to express its magnitude, it is invalid.

∴ +8 needs 5-bit, 01000 (+ve)

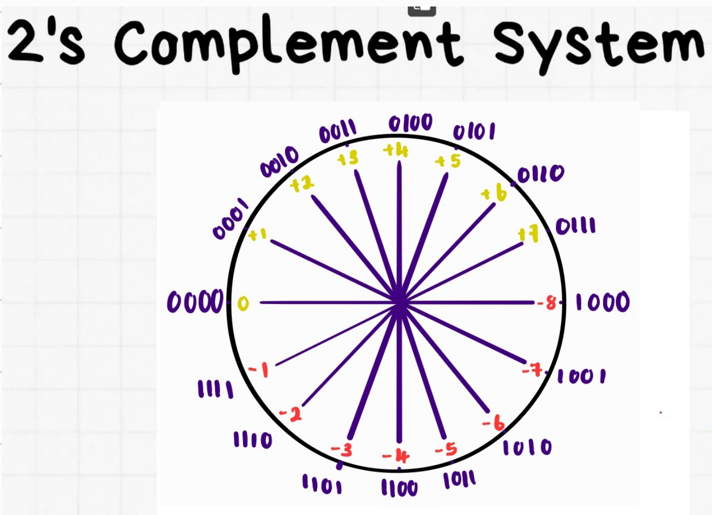


Range -2^{n-1} to $(2^{n-1} - 1)$

② 2's compliment

+ve
+6 = 0 1 1 0
↓ inverse
-ve
(1's) -6 = 1 0 0 1
↓ +1
-ve
(2's) -6 = 1 0 1 0

+ve
+7 = 0 1 1 1
↓ inverse
-ve
(1's) -7 = 1 0 0 0
↓ +1
-ve
(2's) -7 = 1 0 0 1



1's Complement Arithmetics

(a) & (b) same as 2's complement

(c) Pos + Neg numbers, Neg > Pos

```
+5    0101
-6    1001
----------
-1    1110 (correct answer)
```

(d) Pos + Neg numbers, Pos > Neg

```
-5    1010
+6    0110
   (1) 0000
          1 (end-around carry)
----------
+1    0001 (correct; no overflow)
```

(e) Negative no's. |sum| <= 2^(n-1)

```
-3    1100
-4    1011
   (1) 0111
          1 (end-around carry)
----------
-7    1000 (correct; no overflow)
```

(f) Negative no's. |sum| > 2^(n-1)

```
-5    1010
-6    1001
   (1) 0011
          1 (end-around carry)
----------
      0100 (wrong answer; overflow)
```

4-bit 1s compliment range is only $-7 < x < 7$

```
−3    1100
−4    1011
   (1) 0111
        └──→ 1   (end-around carry)
       1000      (correct answer, no overflow)
```

```
−5    1010
−6    1001
   (1) 0011
        └──→ 1   (end-around carry)
       0100      (wrong answer because of overflow)
```

2's Complement Arithmetics

```
  4     0100
+ 3     0011
  7     0111
```

```
  −4       1100
+ (−3)     1101
  −7      11001
```

Ignore overflow

```
  4     0100
− 3     1101
  1    10001
```

Ignore overflow

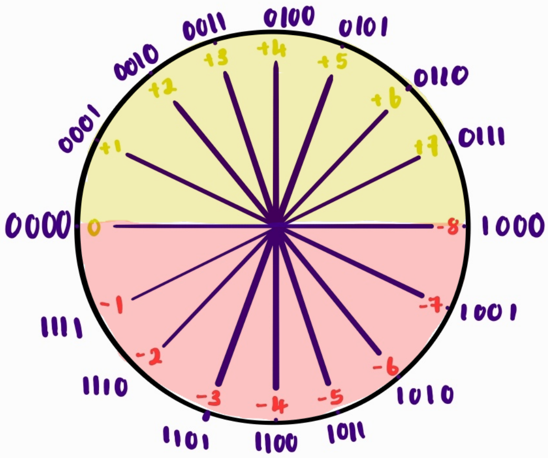
```
 −4     1100
+ 3     0011
 −1     1111
```

The concept behind 1's & 2's compliments

2's compliment

Draw a circle with ascending binary numbers clockwise & assign the signed binary numbers according to 2's complement conversion

Example 4 bit 2's compliment



Take note that to go to -2 from 0, there are 2 ways

1) Go backwards (counter-CW) by 2 steps = -2
2) Go forward (CW) by [No. of # - 2] = 16 - 2 = 14

2^n ← no. of bits

General formula to go forward and get a negative number k

$-k = 2^n - k$

$= (2^n - 1) + 1 - k$

$= \boxed{\text{(All ones in n-bits)} - k} + 1$

$\boxed{-k = \text{Invert of } k + 1 \quad \#}$

∴ If I want -3 in 2's complement [4 bit]

$\Longrightarrow -3 = [1111 - 0011] + 1$

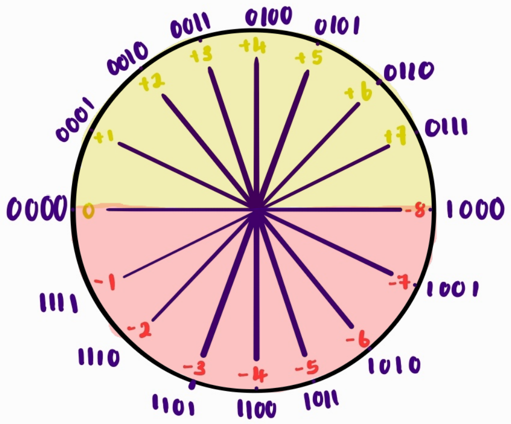
$= 1100 + 1$

$-3 = 1101$ #

∴ Two's complement encodes a negative number as the forward distance from 0 around the n-bit clock in binary

Addition & substraction

Before going into this topic, take note of the following



What would 10000 represent in this 4-bit 2's complement?

1) $\overset{2^4}{1}\ \overset{2^3}{0}\ \overset{2^2}{0}\ \overset{2^1}{0}\ \overset{2^0}{0}$ represents taking 16 steps forward

2) But since a 4-bit number also has 16 numbers, taking 16 steps forward would mean taking 1 whole circle / 360°

3) Thus in this 4-bit, 2's complement, $x + 10000 = x$

∴ Thus for a n-bit number, the n+1 bit [2^n] can be ignored.

Example 1

$$\begin{array}{rr} {}^{1}0010 & 2 \\ +\ 0011 & 3 \\ \hline 0101 & 5 \\ \hline \end{array}$$

∴ Interpret this as, if I were to take 0011 [3] steps ~~cw~~ forward from 0010 [2], I'll reach 0101 [5]

Example 2

$$\begin{array}{rr} {}^{1}0\,1\,{}^{1}0\,1 & 5 \\ +\ 1\,1\,0\,1 & +\ (-3) \\ \hline \boxed{1}\,0\,0\,1\,0 & 2 \\ \hline \end{array}$$

$= 0010$

$= 2$

This is called an carry-out

[This just means walking a full revolution]

Since this doesn't change the value, we ignore it & only look at the first 4 bits

Overflow [Out of range]

$$\begin{array}{r} 0\,1\,0\,1 \\ +\ 0\,1\,0\,0 \\ \hline \boxed{1}\,0\,0\,0 \\ \hline \end{array}$$

carry can't be on the sign bit

OR

$$\begin{array}{rr} \overset{-ve}{1}\,0\,0\,1 & -7 \\ +\ \overset{-ve}{1}\,0\,1\,0 & +\ (-6) \\ \hline 1\,\overset{+ve}{0}\,0\,1\,1 & 3\,? \\ \hline \end{array}$$

x

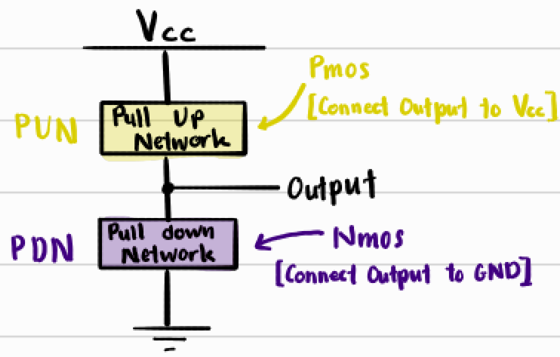
Let's say we ignore

-ve plus -ve won't be +ve.

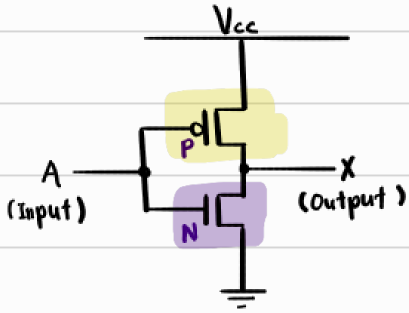
CMOS Logic Gate

Pmos on when 0
Nmos on when 1

Structure of CMOS

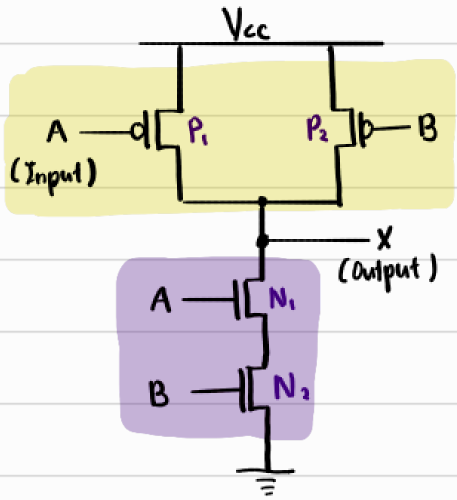


Not gate [$\overline{A}$]



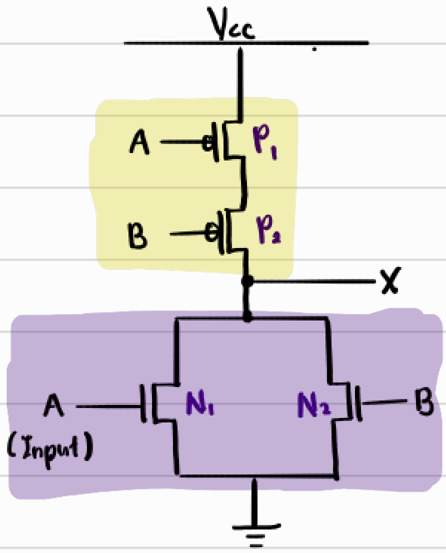
A	P	N	X
1	0	1	0
0	1	0	1

NAND gate [$\overline{A \cdot B}$]



A	P_1	N_1	B	P_2	N_2	X
0	1	0	0	1	0	1
1	0	1	0	1	0	1
0	1	0	1	0	1	1
1	0	1	1	0	1	0

NOR gate [$\overline{A + B}$]



A	P_1	N_1	B	P_2	N_2	X
0	1	0	0	1	0	1
1	0	1	0	1	0	0
0	1	0	1	0	1	0
1	0	1	1	0	1	0

Conclusion

NAND ~ PUN Parallel
PDN Series

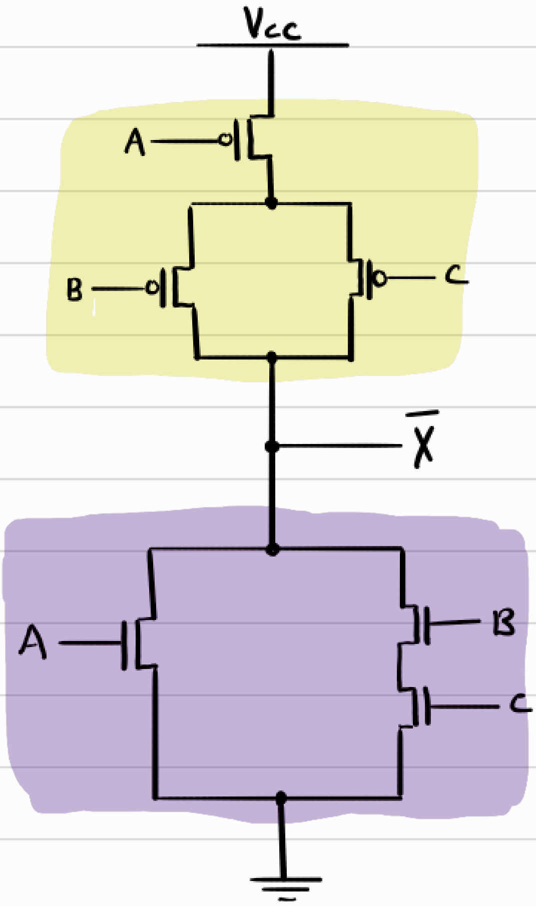
NOR ~ PUN Series
PDN Parallel

Example 1

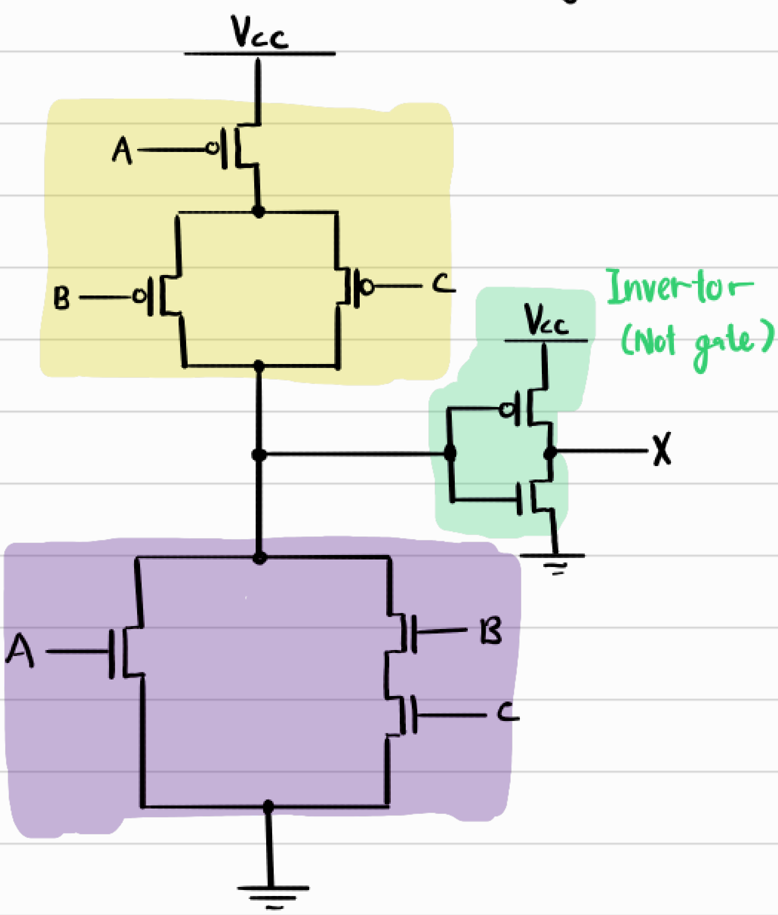
Construct a CMOS Logic Gate whose out is $X = A + (B \cdot C)$

NAND $(\cdot)$	~	PUN Parallel
		PDN Series
NOR $(+)$	~	PUN Series
		PDN Parallel

① Do $\overline{X} = \overline{A + (B \cdot C)}$



② Invert $\overline{X}$ to X with NOT gate



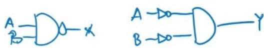
Boolean Algebra

Boolean algebra

NOT: $X = A'$ $\quad \bar{A}$ $\quad \sim A$

AND: $A \cdot B$ $\quad AB$

OR: $A + B$



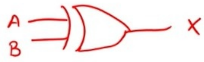
NAND: $\overline{(A \cdot B)}$ $\quad \overline{AB}$ $\quad \overline{AB} \neq \bar{A}\bar{B}$ $\quad X \neq Y$

NOR: $\overline{A+B}$ $\quad \overline{(A+B)}$ $\quad \bar{A} + \bar{B} \neq \overline{A+B}$

Other logic gates

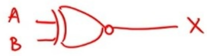
XOR / XNAND

$X = A \oplus B$



XNOR / XAND

$X = \overline{A \oplus B}$



Theorems of Boolean algebra

$\boxed{\text{input} = X}$

(1) Identity theorem

$X + 0 = X \qquad X \cdot 1 = X$

(2) Null theorem

$X + 1 = 1 \qquad X \cdot 0 = 0$

(3) Idempotent theorem

$X + X = X \qquad X \cdot X = X$

(4) Involution theorem

$\bar{\bar{X}} = X \qquad \overline{(X')} = X \qquad (X')' = X \qquad \sim\sim X = X$

(5) Complementarity theorem

$X + \bar{X} = 1 \qquad X \cdot \bar{X} = 0$

(6) Commutativity theorem

$X + Y = Y + X \qquad XY = YX$

(7) Associativity theorem

$(X+Y) + Z = X + (Y+Z) \qquad (X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$

(8) Distributivity theorem

$X \cdot (Y+Z) = (XY) + (XZ) \qquad X + (Y \cdot Z) = (X+Y) \cdot (X+Z)$

(9) Uniting theorem

$X \cdot Y + X \cdot \bar{Y} = X \qquad (X+Y) \cdot (X + \bar{Y}) = X$

(10 & 11) Absorption theorem

$X + XY = X \qquad X \cdot (X+Y) = X$

$(X + \bar{Y}) \cdot Y = X \cdot Y \qquad (X \cdot \bar{Y}) + Y = X + Y$

(12) Factoring theorem

$(X+Y) \cdot (\bar{X} + Z) = XZ + \bar{X}Y \qquad XY + \bar{X}Z = (X+Z) \cdot (\bar{X} + Y)$

(13) Consensus theorem

$(XY) + (YZ) + (\bar{X}Z) = XY + \bar{X}Z$

$(X+Y) \cdot (Y+Z) \cdot (\bar{X}+Z) = (X+Y) \cdot (\bar{X}+Z)$

DeMorgan's Law

$\overline{X+Y} = \bar{X} \cdot \bar{Y}$

$\overline{X \cdot Y} = \bar{X} + \bar{Y}$

General form

$\overline{(X_0 + X_1 + X_2 + \cdots)} = \bar{X}_0 \cdot \bar{X}_1 \cdot \bar{X}_2 \cdot \cdots$

Allowed Operations

- Commutativity ~ $X + Y = Y + X$
  ~ $XY = YX$
- Associativity ~ $(X + Y) + Z = X + (Y + Z)$
  ~ $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$
- Distributivity ~ $X(Y + Z) = XY + XZ$
  ~ $X + (Y \cdot Z) = (X + Y) \cdot (X + Z)$
- Factoring ~ $(A + B)(C + D) = AC + AD + BC + BD$

  Special case : $(X + Y) \cdot (X' + Z) = XZ + X'Y$

Theorems

OR (+)	Theorem	And (·)
$X + 0 = X$	Identity	$X \cdot 1 = X$
$X + 1 = 1$	Null	$X \cdot 0 = 0$
$X + X = X$	Idempotent	$X \cdot X = X$
$\overline{\overline{X}} = X$	Involution	$\overline{\overline{X}} = X$
$X + \overline{X} = 1$	Complementarity	$X \cdot \overline{X} = 0$
$(X \cdot Y) + (X \cdot \overline{Y}) = X$	Uniting	$(X + Y) \cdot (X + \overline{Y}) = X$
$X + XY = X$	Absorbtion	$(X \cdot \overline{Y}) + Y = X + Y$
$XY + YZ + \overline{X}Z = XY + \overline{X}Z$	Consensus	$(X + Y) \cdot (Y + Z) \cdot (\overline{X} + Z) = (X + Y) \cdot (\overline{X} + Z)$
$\overline{X + Y} = \overline{X} \cdot \overline{Y}$	DeMorgan	$\overline{XY} = \overline{X} + \overline{Y}$